



# College Management System

*Transforming Traditional Campuses into AI-Enabled Smart Campuses.*

*"Empowering minds, automating systems, and building the future of campus governance."*

*Why manage when you can innovate? Experience the ultimate synergy of AI, analytics, and automation.*

The Future of Education Starts  
Here  
AI \* Automation \* Analytics \*  
Intelligence

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# 1. ADMISSION MANAGEMENT

## 1. Key Sub-Modules

Online Registration | Entrance/Management Quota | Document Verification | Fee Payment | Seat Allotment | Student ID Generation | Department Allocation

## 2. Operational Features

- Interactive student portal with validation checks.
- Rank merit list calculations and fee status verification.
- Automatic student profile and institutional ID card initialization.

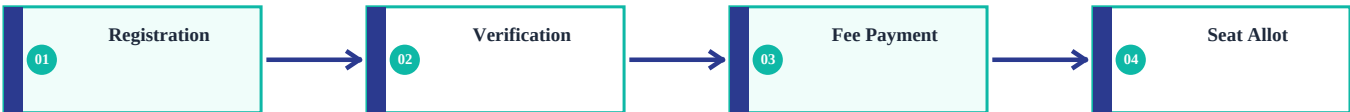
## 3. Working Process Flow

Prospects register online. The system checks cutoffs, schedules document verification slots, records quota allocations, and locks seats upon fee transaction confirmation. Finally, it allocates department branches and releases roll numbers.

## 4. Role-Based Access Controls (RBAC)

| Authorized Role      | Access Scope & Permissions Matrix                                                |
|----------------------|----------------------------------------------------------------------------------|
| Super Admin / Admins | Complete control. Modify fee status, override quota, edit document verify rules. |
| Principal / Dean     | Approve final seat allocation, check merit logs and intake stats.                |
| Student / Guest      | Fill application forms, upload verification files, pay fees, download receipts.  |

## 5. System Operational Workflow Flowchart



## 2. STUDENT INFORMATION SYSTEM (SIS)

### 1. Key Sub-Modules

Student Profile | Academic History | Certificates | Parent Details | ID Cards | Alumni Records

### 2. Operational Features

- Central 360-degree digital card for every registered student.
- Automated certificate generation (Bonafide, Transfer, Course Completion).
- Continuous tracking transitioning to alumni database on graduation.

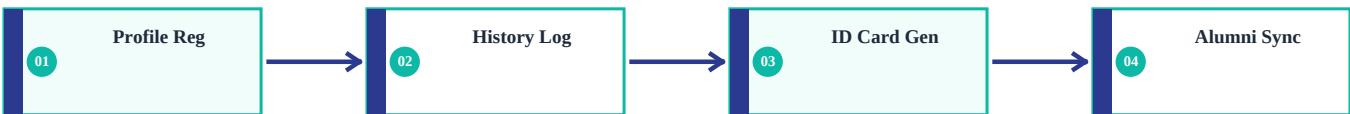
### 3. Working Process Flow

Students input details and upload files. System formats ID cards, links course completion records, logs discipline logs, and manages parent contacts. Upon graduation, accounts transfer into the alumni directory automatically.

### 4. Role-Based Access Controls (RBAC)

| Authorized Role      | Access Scope & Permissions Matrix                                            |
|----------------------|------------------------------------------------------------------------------|
| Super Admin / Admins | Full read/write rights. Manage certificate formats and disciplinary records. |
| HOD / Faculty        | View student profile dashboards, check academic progress and parent details. |
| Student / Parents    | Read-only access to personal profiles and grade cards. Update address logs.  |

### 5. System Operational Workflow Flowchart



### 3. ACADEMIC MANAGEMENT

#### 1. Key Sub-Modules

Academic Calendar | Semester Creation | Course Registration | Curriculum (R20/R23 etc.) | Credit System | Electives | Timetable

#### 2. Operational Features

- Configurable curriculum regulations (Choice Based Credit System - CBCS).
- Automatic elective courses allocation based on student preference and merit.
- Clash-free timetabling engines mapping classrooms and faculty availability.

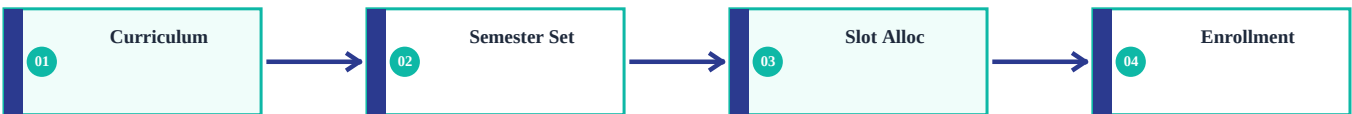
#### 3. Working Process Flow

Deans create semesters and define course codes under specific regulations (e.g. R23). Faculty input lecture availability. System builds visual calendar timetables. Students login to select electives and register courses.

#### 4. Role-Based Access Controls (RBAC)

| Authorized Role     | Access Scope & Permissions Matrix                                              |
|---------------------|--------------------------------------------------------------------------------|
| Super Admin / Deans | Create semesters, regulations, map core timetables and publish calendar books. |
| HOD / Faculty       | Add syllabus, verify curriculum files, manage course mappings.                 |
| Student             | Select open electives and complete course registrations.                       |

#### 5. System Operational Workflow Flowchart



4. ATTENDANCE

1. Key Sub-Modules

Faculty Attendance Entry | Student Attendance | Biometric/RFID Integration | Shortage Calculation | Parent SMS | Reports

2. Operational Features

- Biometric device integration pulling punch logs in real time.
- Automated attendance shortage reports tracking minimum 75% thresholds.
- SMS triggers dispatching daily warnings to parents for absent students.

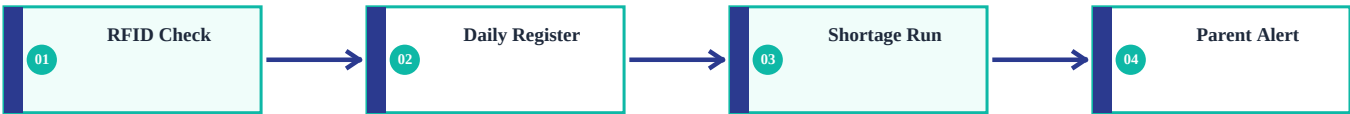
3. Working Process Flow

Biometric devices log check-ins. Faculty verify records manually or override errors. The shortage calculator computes compliance scores weekly and schedules SMS alerts. HODs export report indices.

4. Role-Based Access Controls (RBAC)

| Authorized Role     | Access Scope & Permissions Matrix                                        |
|---------------------|--------------------------------------------------------------------------|
| Super Admin / Admin | Configure biometric ports, clear buffer errors, override shortage lists. |
| HOD / Faculty       | Submit daily class attendance sheets, review department percentages.     |
| Student / Parent    | Read-only check-in calendar logs and warning notifications.              |

5. System Operational Workflow Flowchart



## 5. EXAMINATION

### 1. Key Sub-Modules

Exam Registration | Hall Tickets | Seating Plan | Internal Marks | External Marks | Grace Marks | Result Processing | SGPA/CGPA | Grade Cards | Revaluation | Supplementary | Transcript

### 2. Operational Features

- Outcome-based marks evaluation with SGPA/CGPA calculations.
- Automated Hall Ticket release blocks for fee defaulters.
- Configurable grade boundaries and grace marks algorithms.

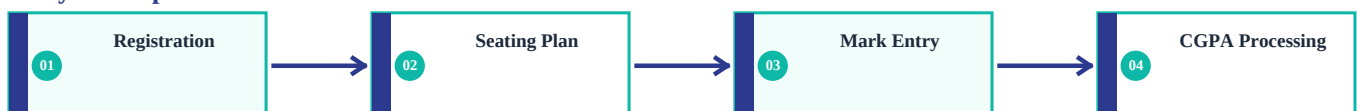
### 3. Working Process Flow

Exam branch sets dates. Students register and system screens backlog limits. Halls and seats are auto-arranged. Faculty upload internal marks. External marks are uploaded via Excel keys. Results process, and transcripts generate.

### 4. Role-Based Access Controls (RBAC)

| Authorized Role               | Access Scope & Permissions Matrix                                                   |
|-------------------------------|-------------------------------------------------------------------------------------|
| Super Admin / Exam Controller | Full administrative control over result processing, grace marks, and grade cards.   |
| HOD / Faculty                 | Upload internal marks, review revaluation applications.                             |
| Student                       | Register for exams, download hall tickets, view results, and apply for revaluation. |

### 5. System Operational Workflow Flowchart



6. FACULTY ERP

1. Key Sub-Modules

Faculty Profile | Workload | Leave | Attendance | Research | Publications | Performance | Payroll

2. Operational Features

- Workload manager tracking weekly lecture schedules against guidelines.
- Research and publication log tracking research citations.
- Self-appraisal sheets linked directly to NAAC files.

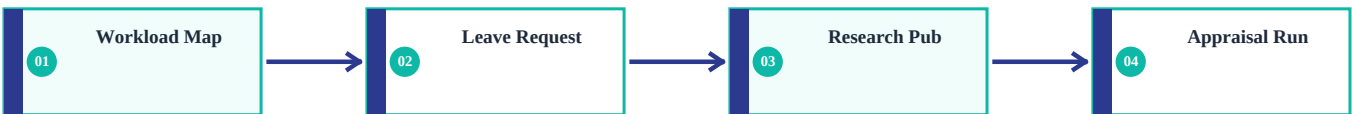
3. Working Process Flow

Faculty edit profiles, upload research papers, apply for leaves, and track weekly lecture workload. System computes performance scores based on publications and feedback, updating payroll allowance indices.

4. Role-Based Access Controls (RBAC)

| Authorized Role     | Access Scope & Permissions Matrix                                        |
|---------------------|--------------------------------------------------------------------------|
| Super Admin / Admin | Approve promotions, modify payroll bands, adjust teaching limits.        |
| HOD / Dean          | Review faculty workloads, approve research funds, sign leaves.           |
| Faculty             | Manage personal profiles, leaves, research logs, and check salary slips. |

5. System Operational Workflow Flowchart



## 7. LEARNING MANAGEMENT

### 1. Key Sub-Modules

[Notes](#) | [Videos](#) | [Assignments](#) | [Quizzes](#) | [Discussion Forum](#) | [Online Classes](#)

### 2. Operational Features

- Direct file uploads (PDF, MP4, Docs) for study material.
- Online assignment submissions with plagiarism detection cues.
- Online quiz modules with automatic grading keys.

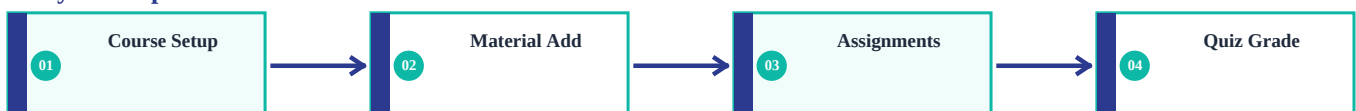
### 3. Working Process Flow

Faculty create course pages and post notes, files, and video links. Quizzes are published with timers. Students submit assignments. Discussion forums allow thread queries. Grades sync to the LMS database.

### 4. Role-Based Access Controls (RBAC)

| Authorized Role | Access Scope & Permissions Matrix                                           |
|-----------------|-----------------------------------------------------------------------------|
| HOD / Faculty   | Upload study material, manage quizzes, grade submissions, run online links. |
| Student         | Download notes, watch videos, submit assignments, attempt quizzes.          |

### 5. System Operational Workflow Flowchart



8. PLACEMENT CELL

1. Key Sub-Modules

Student Eligibility | Company Registration | Job Posting | Applications | Aptitude Tests | Interview Schedule | Offer Letters | Placement Statistics

2. Operational Features

- Eligibility criteria filters (CGPA, backlogs limits).
- Recruiter dashboard to post vacancies and download applicant profiles.
- Placement analytics tracking average and highest package values.

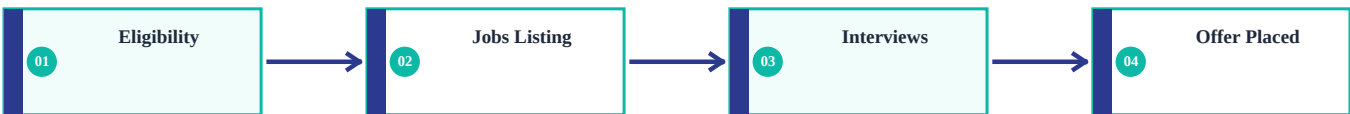
3. Working Process Flow

Placement officer registers companies. Recruiters upload jobs. Eligible students register. System arranges aptitude exams and interview panels. Offers are logged, updating placement dashboard statistics.

4. Role-Based Access Controls (RBAC)

| Authorized Role         | Access Scope & Permissions Matrix                                                |
|-------------------------|----------------------------------------------------------------------------------|
| Placement Officer (TPO) | Add corporate profiles, review applications, set CGPA thresholds, upload offers. |
| Recruiter               | Post vacancies, filter resumes, update hiring status.                            |
| Student                 | View jobs, check eligibility status, apply, and download mock tests.             |

5. System Operational Workflow Flowchart



9. HOSTEL

1. Key Sub-Modules

Room Allocation | Attendance | Hostel Fees | Visitor Log | Complaints

2. Operational Features

- Live bed matrix displaying vacant, occupied, and maintenance room layouts.
- Outing gate logs and night check-in attendance registries.
- Complaints ticket portal routing issues to wardens.

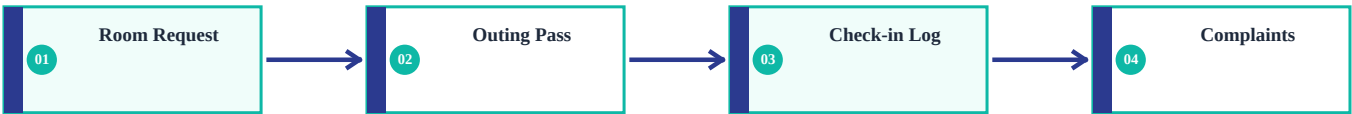
3. Working Process Flow

Students apply for rooms. System checks fees and assigns beds. Wardens take night attendance via mobile keys. Security logs visitors. Mess systems track student feedbacks.

4. Role-Based Access Controls (RBAC)

| Authorized Role  | Access Scope & Permissions Matrix                                              |
|------------------|--------------------------------------------------------------------------------|
| Hostel Warden    | Allocate rooms, verify night check-ins, resolve maintenance complaints.        |
| Accounts Manager | Post hostel fees and track pending payments.                                   |
| Student Resident | Apply for rooms, submit gatepasses, register complaints, log dining feedbacks. |

5. System Operational Workflow Flowchart



10. TRANSPORT

1. Key Sub-Modules

Route Management | Bus Allocation | GPS Tracking | Driver Details | Fee Collection

2. Operational Features

- Route mapper tracking stops and bus seating capacities.
- GPS location integration for live bus coordinates.
- Fee collection registers linked to student entry passes.

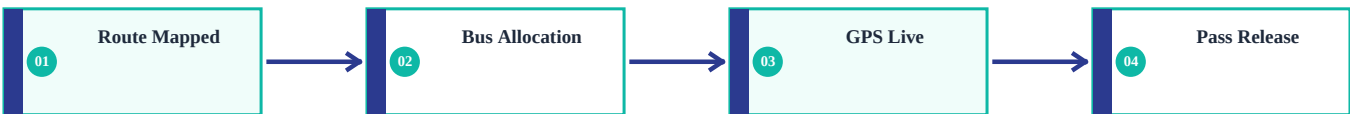
3. Working Process Flow

Transport in-charge defines routes and stops. Buses are allocated. Drivers register. Students opt for routes, pay fees, and get boarding cards. GPS devices feed coordinates to the student dashboard.

4. Role-Based Access Controls (RBAC)

| Authorized Role   | Access Scope & Permissions Matrix                                    |
|-------------------|----------------------------------------------------------------------|
| Transport Officer | Define routes, schedule maintenance, allocate drivers, track routes. |
| Accounts Cell     | Manage fee logs and bus pass issue lists.                            |
| Student / Parent  | Read-only access to route schedules, bus locations, and fee status.  |

5. System Operational Workflow Flowchart



11. LIBRARY

1. Key Sub-Modules

Book Catalog | Issue/Return | Fine | Digital Library | Barcode/RFID

2. Operational Features

- OPAC Search index for physical book titles and journals.
- RFID/Barcode tracking scanning book issuance and returns.
- Automated library fine calculations linked to fee books.

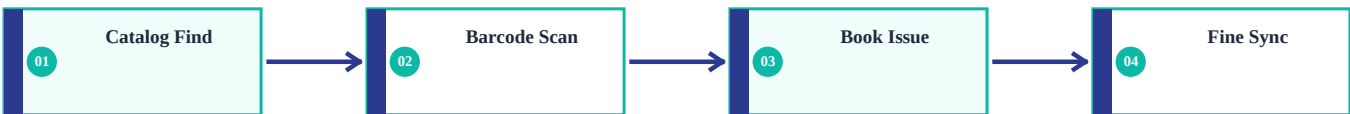
3. Working Process Flow

Librarian scans book barcodes. System logs check-out limits and return periods. Reminders are dispatched for overdue items. Fine calculations are linked to student accounts automatically.

4. Role-Based Access Controls (RBAC)

| Authorized Role   | Access Scope & Permissions Matrix                                           |
|-------------------|-----------------------------------------------------------------------------|
| Librarian         | Manage catalog, scan barcodes, override fines, issue digital books.         |
| Student / Faculty | Search catalog, view checkout history, read e-books, pay outstanding fines. |

5. System Operational Workflow Flowchart



12. FINANCE

1. Key Sub-Modules

Tuition Fees | Scholarships | Payroll | Vendor Payments | GST Reports | Accounting

2. Operational Features

- Tuition fee ledger tracking dues, installment plans, and online gateways.
- Scholarship adjustment registers and government scheme mappings.
- TDS, GST, and balance sheet generators.

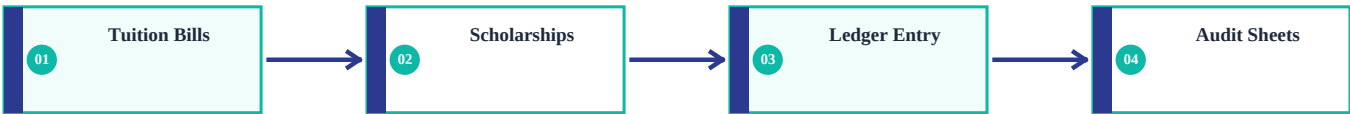
3. Working Process Flow

Finance team logs fees, checks scholarship allocations, dispatches vendor payments, and runs payroll accounts. GST records and balance sheets are compiled automatically for annual audit reporting.

4. Role-Based Access Controls (RBAC)

| Authorized Role            | Access Scope & Permissions Matrix                                           |
|----------------------------|-----------------------------------------------------------------------------|
| Super Admin / Finance Head | Full financial control, ledger edits, balance sheets, bank reconciliations. |
| Accounts Team              | Process tuition fees, verify scholarships, release vendor bills.            |
| Student / Parent           | View fee balances, pay online, download receipts.                           |

5. System Operational Workflow Flowchart



13. HRMS

1. Key Sub-Modules

Employee Records | Recruitment | Attendance | Leave | Payroll | Performance

2. Operational Features

- Staff directory with reactive details.
- Leave management board and biometric sync controllers.
- Locked payroll disbursement keys.

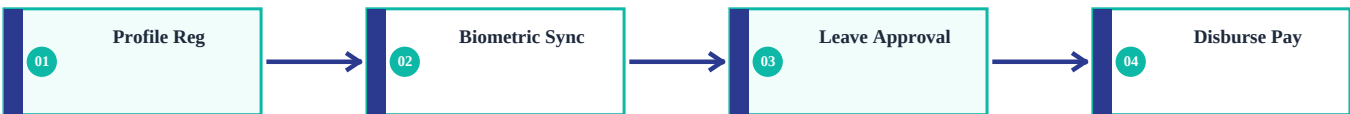
3. Working Process Flow

Admin manages personnel records. Department heads sync RFID devices and approve leaves. System calculates basic scales, allowance, and releases salaries. Appraisals track research files for NAAC audits.

4. Role-Based Access Controls (RBAC)

| Authorized Role      | Access Scope & Permissions Matrix                               |
|----------------------|-----------------------------------------------------------------|
| Super Admin / Admins | Manage records, post openings, approve leaves, release payroll. |
| Accounts Manager     | Disburse salary scales, edit allowances.                        |
| HOD / Faculty        | Approve leaves, sync attendance, view performance ratings.      |

5. System Operational Workflow Flowchart



14. INVENTORY

1. Key Sub-Modules

Assets | Lab Equipment | Purchase Orders | Stock | Maintenance

2. Operational Features

- Book Value and WDV calculations with depreciation rates.
- Stock consumable indicators warning on low volumes.
- Outward gatepass log with security return checks.

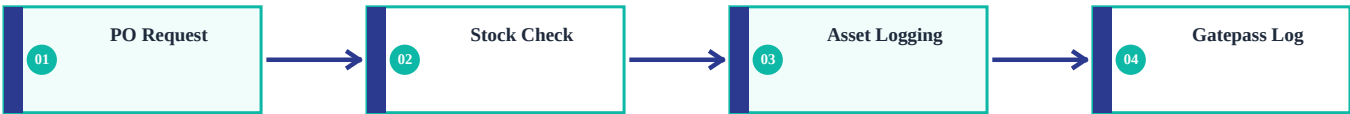
3. Working Process Flow

Asset register audits initial values vs. depreciation values. Custodians log lab equipment states. Purchase requests are authorized. Stock levels auto-adjust. Gate passes track devices sent out for repair.

4. Role-Based Access Controls (RBAC)

| Authorized Role     | Access Scope & Permissions Matrix                            |
|---------------------|--------------------------------------------------------------|
| Super Admin / Admin | Full asset ledger access, PO approvals, gatepass issuing.    |
| Accounts            | Verify PO budgets, balance asset books, restock consumables. |
| HOD / Lab Custodian | Mark equipment serviced, request gatepasses.                 |

5. System Operational Workflow Flowchart



15. ACCREDITATION

1. Key Sub-Modules

NBA | NAAC | AICTE | NIRF | AISHE Reports | Event Management | Certificate Management | Evidence Repository

2. Operational Features

- NBA CO-PO mapping and attainment | NAAC SSR criteria-wise evidence.
- AICTE EOA records, NIRF data, and AISHE DCF filing compliance.
- Event proposal workflow, attendee check-in, and auto-certificates.
- Digital signed certificate downloads and central evidence repositories.

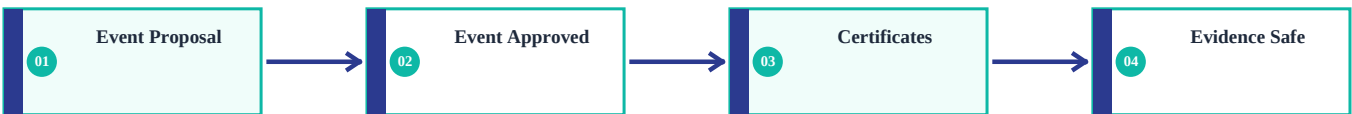
3. Working Process Flow

Faculty/HOD creates an event proposal. The HOD and Principal review and approve it. The event is published for student registration, attendance is verified, and the coordinator uploads the reports and documents. The system then automatically generates participation/organizer certificates and stores event evidence for accreditation criteria.

4. Role-Based Access Controls (RBAC)

| Authorized Role       | Access Scope & Permissions Matrix                                                          |
|-----------------------|--------------------------------------------------------------------------------------------|
| Super Admin           | Manage accreditation modules, approve reports, manage certificates, view analytics.        |
| Principal / Dean      | Approve events, verify accreditation reports, approve final submissions, monitor progress. |
| HOD                   | Create/approve event proposals, upload department evidence, validate reports.              |
| Faculty / Coordinator | Organize events, mark attendance, upload reports, generate certificates.                   |
| Student               | Register for events, view approvals, download certificates, track participation history.   |

5. System Operational Workflow Flowchart



16. COMMUNICATION

1. Key Sub-Modules

SMS | Email | Push Notifications | Circulars | WhatsApp

2. Operational Features

- Multi-channel broadcast engine routing alerts via SMS, Email, and WhatsApp.
- Circular archive for official notices.
- Automated triggers for announcements, fee dues, and exam updates.

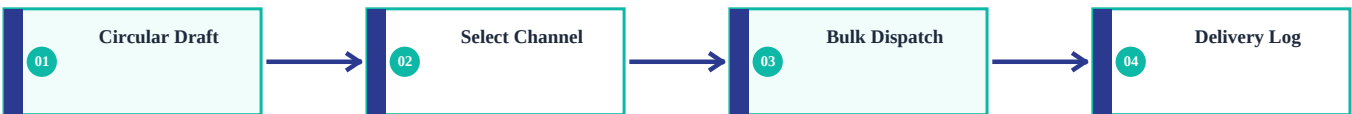
3. Working Process Flow

Authorized staff select channels and enter messages. The communication dispatcher sends bulk messages. Push notifications appear on mobile devices, and official circulars publish to dashboards.

4. Role-Based Access Controls (RBAC)

| Authorized Role      | Access Scope & Permissions Matrix                   |
|----------------------|-----------------------------------------------------|
| Super Admin / Admins | Send global notifications, upload public circulars. |
| HOD / Faculty        | Message department students, issue class circulars. |
| Student / Parents    | Read-only notifications inbox.                      |

5. System Operational Workflow Flowchart



## 17. GRIEVANCE

### 1. Key Sub-Modules

Complaints | Anti-Ragging | Women Cell | Resolution Tracking

### 2. Operational Features

- Anonymous grievance submission portals for security.
- Legal compliance trackers for Anti-Ragging and Women Cell cases.
- Resolution tickets routing incidents to designated review boards.

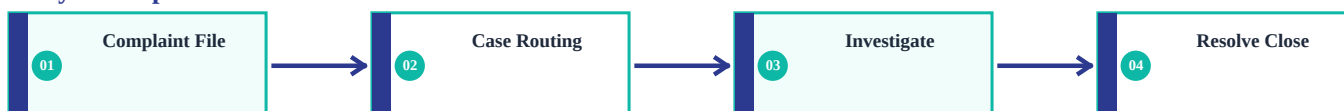
### 3. Working Process Flow

Grievances are registered. The system assigns a case number and notifications are sent to the review board. Investigation logs are filed, and status shifts to Resolved after review.

### 4. Role-Based Access Controls (RBAC)

| Authorized Role     | Access Scope & Permissions Matrix                           |
|---------------------|-------------------------------------------------------------|
| Grievance Committee | View reports, assign investigators, record resolution logs. |
| Super Admin         | Oversee case statistics and check compliance.               |
| Student / Faculty   | Submit complaints, track ticket status anonymously.         |

### 5. System Operational Workflow Flowchart



18. ALUMNI

1. Key Sub-Modules

Alumni Registration | Donations | Events | Mentorship

2. Operational Features

- Verified alumni registry matching student graduation records.
- Secure donation gateways with tax benefit receipts.
- Mentorship network pairing alumni with current students.

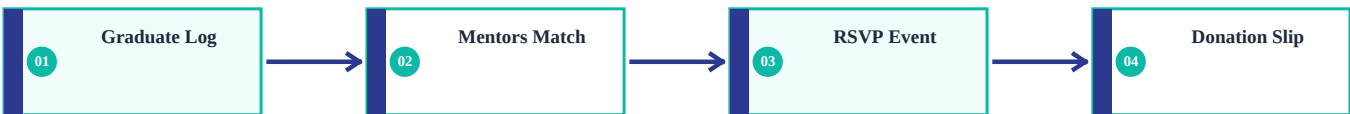
3. Working Process Flow

Graduating students transfer to the alumni directory. Alumni update details, RSVP to events, select mentorship tracks, and process donations. System generates 80G tax benefit certificates.

4. Role-Based Access Controls (RBAC)

| Authorized Role        | Access Scope & Permissions Matrix                                      |
|------------------------|------------------------------------------------------------------------|
| Alumni Officer / Admin | Verify registry profiles, coordinate events, monitor donations.        |
| Alumni                 | Manage profile details, process contributions, offer mentorship slots. |
| Student                | Request mentorship matches, view alumni networks.                      |

5. System Operational Workflow Flowchart



19. ADMINISTRATION

1. Key Sub-Modules

User Roles | Permissions | Audit Logs | Backup | Reports | Analytics Dashboard

2. Operational Features

- Granular role permissions configuration tool.
- Immutable audit logs detailing every user action.
- System database backup schedules.

3. Working Process Flow

Super admins use the dashboard to map user credentials to roles. The system records all API calls in an immutable audit log. Automated cron-jobs run database backups nightly.

4. Role-Based Access Controls (RBAC)

| Authorized Role  | Access Scope & Permissions Matrix                                                  |
|------------------|------------------------------------------------------------------------------------|
| Super Admin      | Manage system configurations, edit permissions, view audit logs, schedule backups. |
| Auditors / Deans | View logs and export analytics report files.                                       |

5. System Operational Workflow Flowchart

